

STANLEY COUNTY AP, NC, USA

WMO: 722148

Lat: 35.417N

Lon: 80.151W

Elev: 609

StdP: 14.38

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 63824

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.5	21.8	-0.1	5.6	28.3	5.4	7.4	30.3	19.6	53.5	17.5	50.2	2.0	350	0.386

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.7	95.5	74.7	92.7	75.0	90.6	74.7	78.8	88.9	77.8	87.6	76.9	86.1	5.8	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
(p)															
76.2	139.7	82.8	75.2	135.0	81.5	74.4	131.5	80.5	42.8	89.3	41.6	87.7	40.6	86.6	85.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
17.0	14.5	12.2	DB	11.1	98.9	4.7	4.2	7.7	101.9	4.9	104.4	2.2	106.8	-1.2	109.9	
			WB	10.2	81.1	4.6	2.1	6.9	82.6	4.2	83.8	1.7	85.0	-1.7	86.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	60.4	40.5	43.7	50.5	59.8	68.1	76.3	79.1	78.1	72.7	60.8	49.8	44.4
	DBStd	15.60	9.62	9.46	9.19	7.56	6.52	4.79	3.81	4.38	5.41	7.84	8.00	8.79
	HDD50	997	319	220	110	12	0	0	0	0	0	9	101	226
	HDD65	3329	759	597	456	187	47	1	0	0	8	173	459	642
	CDD50	4798	25	45	125	307	562	790	901	872	681	345	94	51
	CDD65	1654	0	2	6	31	144	341	436	407	239	44	2	2
	CDH74	15498	0	16	104	446	1362	3312	4148	3693	1981	406	24	6
	CDH80	6420	0	0	11	80	438	1448	1919	1660	776	88	0	0
(14) Wind	WSAvg	4.1	4.3	4.9	5.4	5.4	4.2	3.5	3.2	3.1	3.9	3.8	3.9	3.8
(15) Precipitation	PrecAvg	48.1	3.9	3.7	4.8	3.3	4.1	4.6	5.0	4.5	4.2	3.4	3.1	3.4
	PrecMax	64.0	10.1	8.2	11.5	7.8	10.0	11.7	16.1	12.6	11.5	16.1	7.4	8.1
	PrecMin	29.3	0.7	0.7	1.2	0.7	1.2	0.1	0.8	0.4	0.0	0.0	0.4	0.4
	PrecStd	7.6	2.1	1.8	2.4	1.7	2.0	2.6	3.4	2.6	2.9	2.9	1.7	1.8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	70.1	76.7	81.3	85.5	91.6	97.4	98.1	100.1	95.1	87.3	77.0	73.0
		MCWB	59.8	63.3	62.0	66.8	72.3	73.9	75.9	72.9	70.7	73.7	63.0	65.4
	2%	DB	65.2	71.4	76.9	82.0	88.4	93.9	95.2	95.1	90.9	81.9	72.6	68.4
		MCWB	58.2	59.5	61.1	65.1	70.9	74.4	76.3	75.5	72.2	68.6	61.7	63.6
	5%	DB	61.4	65.7	72.8	79.4	84.7	91.0	91.8	91.1	88.5	78.6	69.7	64.2
		MCWB	55.3	57.1	58.7	62.8	70.4	74.3	76.0	76.0	72.5	66.2	61.2	60.3
	10%	DB	56.9	61.5	67.8	75.8	81.7	88.6	90.1	89.3	84.3	75.0	64.7	60.7
		MCWB	51.6	52.5	56.9	62.3	68.6	73.6	75.5	75.5	71.5	66.7	57.4	54.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	64.2	65.7	66.4	72.0	75.4	79.2	81.0	80.3	77.6	75.1	69.6	69.6
		MCDB	66.6	73.3	75.5	79.3	84.4	89.3	92.1	90.8	85.6	80.9	71.1	70.3
	2%	WB	60.4	62.7	64.0	69.1	73.4	77.5	79.0	78.5	76.2	73.3	66.0	65.0
		MCDB	63.8	67.7	72.7	76.4	82.1	88.2	89.4	88.3	84.4	77.9	70.3	66.9
	5%	WB	56.9	59.2	61.9	67.2	72.2	76.3	78.0	77.6	74.9	70.9	63.1	61.5
		MCDB	59.9	65.0	68.4	75.0	81.1	86.3	88.1	86.7	82.7	76.0	66.2	64.3
	10%	WB	52.4	54.5	59.4	65.1	70.9	75.2	77.0	76.6	73.4	68.0	59.4	56.2
		MCDB	55.6	59.1	66.2	73.0	79.0	84.4	86.5	85.0	79.9	73.1	64.6	58.9
(35) Mean Daily Temperature Range	5% DB	MDBR	21.0	22.6	24.1	24.9	22.7	21.5	20.7	20.0	20.8	22.5	24.1	20.9
		MCDBR	26.3	29.1	29.8	29.4	26.6	25.3	24.0	23.9	26.5	26.5	28.3	23.7
		MCWBR	19.6	18.6	16.0	14.7	11.4	9.2	8.5	9.4	10.6	13.8	18.3	17.3
	5% WB	MCDBR	21.6	25.2	24.9	23.4	21.6	21.7	20.9	20.5	20.5	19.7	20.6	19.5
		MCWBR	19.1	18.8	15.8	13.8	10.6	9.1	8.5	8.8	9.3	12.3	16.6	16.2
(40) Clear-Sky Solar Irradiance	taub		0.311	0.323	0.358	0.399	0.455	0.484	0.535	0.513	0.425	0.382	0.327	0.309
	taud		2.507	2.469	2.402	2.284	2.167	2.123	2.006	2.052	2.290	2.385	2.498	2.545
	Ebn at Noon		284	292	289	281	266	257	243	246	264	267	275	278
	Edh at Noon		26	30	35	42	48	50	56	52	39	33	26	24
(44) All-Sky Solar Radiation	RadAvg		829	1036	1376	1724	1914	2042	1912	1741	1468	1201	913	714
(45)	RadStd		52	126	99	131	155	148	124	82	129	155	73	62

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (69 neighbors)	+0.83	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+283	+209

Nomenclature: See separate page